Finite Element Analysis of a Galvanic Human Body Communication Channel with a Wrist-Worn Transmitter and Grounded Receiver

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***Abstract*—A major challenge in designing systems that support Human Body Communication (HBC) is to determine the optimal size and placement of transceiver electrodes. Physical experiments to determine optimal configurations are possible but are often time-consuming, costly, and, in many cases, not repeatable. A more practical approach is to perform simulation studies. In this work, we present a Finite Element Method (FEM)-based simulation framework to study the galvanic HBC configuration comprising a wrist-worn transmitter and a grounded receiver, a configuration not well studied in the literature. We developed a custom HBC model for this purpose. The communication channel is modelled using geometries and materials representing the anatomical layers of the human body. Simulation results indicate that electrode geometry and positioning are critical factors in determining HBC performance. Channel gain was found to depends on several parameters: the width of the transmitter electrode (TxW), the gap between transmitter electrodes (TxG), the distance from the transmitter electrodes to the end of the arm (TxD), the alignment offset of the receiver ground electrode (RxA), and the vertical height of the receiver ground electrode (RxH). We observe that increasing TxD, RxH, and RxA leads to higher channel gain, while a larger TxG reduces the gain. Interestingly, increasing TxW initially decreases the gain but, beyond a certain point, increases it again. We also propose a simplified circuit model to explain these observed trends.**

***Index Terms*—Human Body Communication (HBC), Finite Element Analysis, Wearable Technology.**

I. Introduction

In many emerging applications of wearable technology, there is a need for on-body wearables to transmit data to a nearby hub. For example, in healthcare applications, a wearable may transfer data related to the patient's health to the hub for post-processing and diagnosis. In secure access applications, wearables may transfer a secret key to the hub to authenticate the individual. Existing methods for transferring data from wearables to the hub mainly utilize radiative communication technologies such as BLE or WiFi. Being radiative, they transmit data over long distances through the air, making them vulnerable to eavesdropping by nearby attackers [1], [2], as shown in Figure 1(a). Moreover, the lack of directionality in these communication methods results in higher power consumption, which limits the operational life of battery-operated wearables.

Human Body Communication (HBC) has emerged as a promising alternative to BLE and WiFi for such applications [3]. HBC utilizes the human body as a conductive channel to transfer data from the wearable to a nearby hub when the user's body is in direct contact with or in close proximity to the hub [4] (see Figure 1(b)). Because the signal remains within or very close to the body, it is difficult for external adversaries to intercept, reducing the risk of eavesdropping. Also, this signal confinement helps reduce energy consumption during communication.

Many different HBC configurations have been proposed in the literature. Figure 1(c) shows a galvanic-type HBC configuration described by Soumick et al [5]. In this setup, two electrodes on a wrist-worn wearable serve as the HBC transmitter, sending sensitive data. At the receiving end, another pair of electrodes picks up the transmitted signals and decodes the data. This HBC configuration supports a range of applications, such as authorizing payments at public kiosks, transferring health data from wearables to hubs, and securely accessing electronic devices.

A significant challenge in the design of the HBC configuration in Figure 1(c) is determining the optimal sizing and placement of electrodes on the wearable and at the receiver. For instance, if the transmitting electrodes are too large, the wearable may become bulky and uncomfortable; if too small, they may not provide sufficient surface area for effective coupling of signals to the body, leading to weak signals picked at the receiver side. Similar trade-offs apply to the electrodes on the receiver side.

A straightforward approach to address this challenge would be to fabricate and experimentally test physical prototypes with different electrode sizes and positions. However, this process is both time-consuming and costly. Practical challenges also arise, signal propagation through the human body is influenced by tissue properties, electrode contact quality, and environmental noise. These factors vary over time, across environments, and between subjects, affecting the repeatability of the results.

A better approach, and the primary contribution of this

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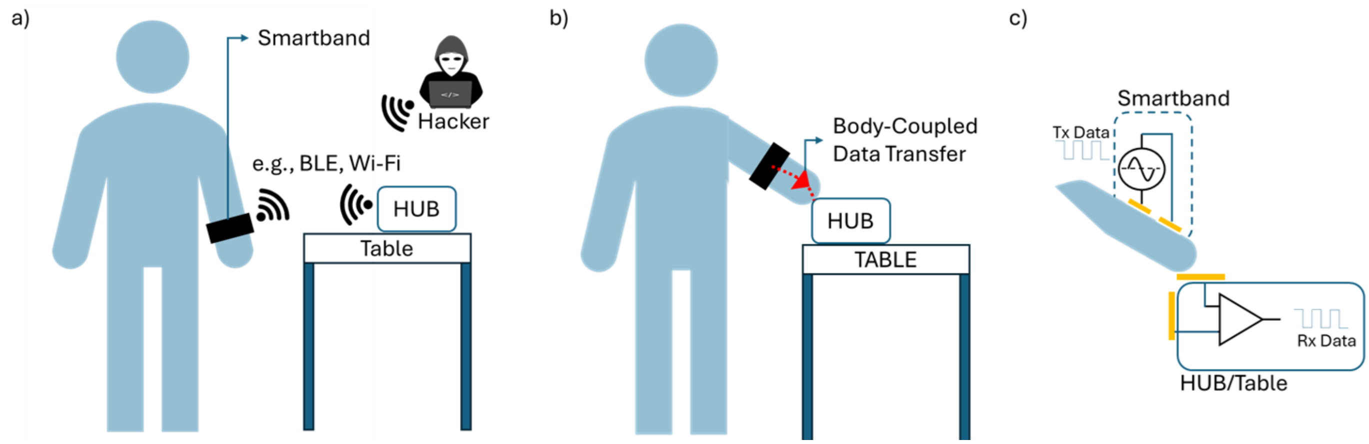


Fig. 1. a) Conventional RF data transfer from a wearable device to a hub, which is vulnerable to eavesdropping. b) Data transfer using Human Body Communication (HBC), where signals are confined within the body and received by the hub only upon physical contact. c) Simplified schematic of the HBC configuration highlighting the transmitter and receiving electrodes.

work, is to perform computer-aided modeling and simulation of the HBC configuration to understand how parameters such as electrode size and position influence HBC characteristics.

Specifically, our contributions are as follows. We develop a simulation-based framework to study the galvanic HBC configuration shown in Figure 1(c). The objective is to analyze how the sizes and positions of the transmitter and receiver electrodes affect overall HBC performance. Our study employs Finite Element Method (FEM) simulations, which are conducted in the electrostatic regime rather than performing full electromagnetic simulations. This simplification is justified, as most HBC applications operate in the sub 1MHz, electrostatic regime, to confine signals within the body. We also developed an HBC model for FEM simulations. The model described in detail in Section III and Section IV-A, incorporates anatomical layers of the human body, including skin, fat, muscle, and bone, along with the transmitting (Tx) and receiving (Rx) electrodes.

The remainder of this paper is organized as follows: Section II reviews related work on FEM-based HBC simulations. Section III describes the system model, including the wearable and receiver design used in FEM simulations. Section IV presents the implementation of the model in Finite Element Method Magnetics (FEMM), a FEM tool [6], along with simulation results. This section also discusses the impact of electrode sizes and their positions on HBC channel quality. Finally, Section V concludes the paper.

II. Related Work

Several studies in the literature focus on FEM simulations of HBC. However, most of these works address wearable-to-wearable communication, where both the transmitter and receiver reside on the subject's body. This configuration is distinct from the HBC setup discussed in this work (Figure 1), which consists of a wearable transmitter and a grounded receiver. Below, we list few such studies.

Yong Song et al. [7], Maity et al. [8] and Wegmueller et al. [9] performed FEM simulations of galvanic HBC, where the human body is modeled in detail using a layered structure consisting of skin, fat, muscle, and bone. Their objective was to determine the dependency of electrode distances and tissue layers on HBC channel characteristics. Muramatsu et al. [10] performed FDTD-based electromagnetic field simulation studies for a galvanic HBC configuration targeting head mounted wearable devices. They focused on understaning the transmission characteristics, power consumption and safety aspects of such HBC configuration to humans. Unlike ours, in these works, the electrodes of both the transmitter and receiver are placed on the body, simulating a wearable-to-wearable communication scenario.

Studies such as those by Muramatsu et al. [11] tried to simplify the human arm into a homogeneous electrical model for FDTD simulations of HBC, but again for wearable-to-wearable configuration.

Hwang et al. [12] conducted FDTD simulations of an HBC configuration to study the dependency of the transmitter's ground electrode on channel characteristics. Tang et al. [13] used FEM simulations to study how body movements and posture affect channel characteristics. Similar to the previous works, these studies also focuses on simulating a wearable-to-wearable communication scenario.

Studies by Fujii et al. [14] and Ruoyu Xu et al. [15] employed FDTD simulations, while Maity et al. [16] and Nath et al. [17] used FEM simulations to study capacitive HBC configurations. Their work differs from ours in two key ways. First, their focus is on wearable-to-wearable communication scenarios. Second, their studies are based on the capacitive mode of HBC, in which only one electrode of both the transmitter and the receiver is in direct contact with the body.

III. System Design

The objective of this work is to study the galvanic HBC configuration shown in Figure 1(c). Specifically, we aim to determine the optimal electrode sizes and placements for improved signal coupling from the transmitter to the receiver through the body. Towards this, we developed a simulation model. Figure 2a) shows this model, which represents a subject wearing a wrist-mounted HBC transmitter that transmits data to a grounded hub in contact.

The human body in the figure models the head, bulk, leg, and arm using rectangular geometries. The dimensions of the model are based on average adult human measurements. The head is designed with dimensions of 11 cm $\times$ 11 cm [18], the torso has a height of 47 cm, and the leg measures 70 cm in length, resulting in an overall model height of approximately 128.4 cm. The arm is modeled with a length of 70 cm [15]. To simplify simulations, the bulk of the body, including the head, torso, and leg, is approximated as a conductor [19].

The arm is modeled in greater detail, since it lies along the HBC signal path. The arm cross-section is built with layers of skin, fat, muscle, and bone tissues to yield more accurate simulation results. The thickness and permittivity values of these layers are provided in Figure 2c) [15] [20].

The HBC transmitter placed on the subject's wrist is modeled using two conducting electrodes: Tx-GND and Tx-Signal. The transmitting signal is applied across these electrodes. At the receiving end, two electrodes are mounted on a wooden table measuring 35 cm $\times$ 108 cm. These electrodes, Rx-GND and Rx-Signal, pick up the HBC signal from the transmitting side, coupled through the subject's body.

Figure 2b) shows the electrical equivalent schematic of the HBC channel. In this schematic, a transmit voltage V_{Tx} is applied across the Tx-Signal and Tx-GND electrodes, and the received voltage V_{Rx} is measured between the Rx-Signal and Rx-GND electrodes, across the capacitive load C_L. The human body and the ambient surroundings represent the communication channel, coupling the signal from the transmitting to the receiving electrodes. Here, the channel gain, defined as V_{Rx}/V_{Tx}, is a function of the geometry and placement of the electrodes. The gain depends on the model parameters TxW (width of the Tx electrode), TxG (gap between Tx electrodes), TxD (distance from the Tx electrodes to the arm end), RxA (alignment offset of the Rx GND electrode), and RxH (vertical height of the Rx GND electrode).

FEM simulations of the model with different values of the parameters can help determine the dependency of TxD, TxG, TxW, RxA, and RxH on the channel gain. This knowledge helps in designing HBC transceivers optimized for compact form factors and stronger signal coupling, leading to improved user experience and data transfer rates. In the next section, we discuss how the model in Figure 2a) is simulated to characterize these dependencies.

IV. Results

In this section, we first discuss the FEM simulation framework, including how the model was constructed and how the simulations were performed. Next, we present the simulation results, highlighting voltage density plots and discussing the dependency of model parameters on channel gain.

A. Simulation Framework

The model was simulated using the Finite Element Method Magnetics (FEMM 4.2) tool [6], with the simulation mode set to electrostatics. Simulations were conducted to determine the voltage distribution across the modeled regions.

A Python script was used to set up the FEM simulations. The script automated the processes of model construction, including the drawing of the 2-D shape, material assignment to different regions, meshing, and boundary condition application. Also, the script was used to run simulations for varying model parameters to analyze their impact on channel gain.

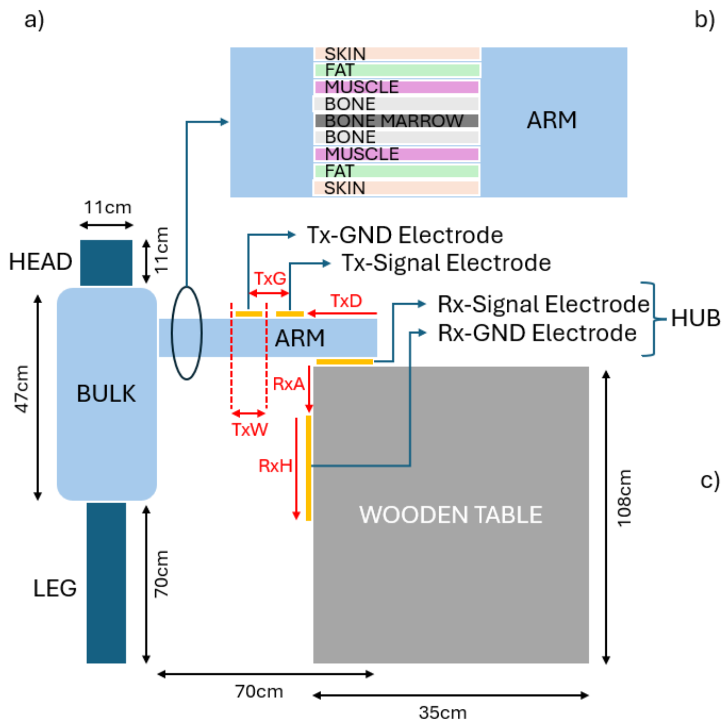
The script was structured as follows. First, the model geometry was constructed using rectangles with appropriate dimensions, as detailed in Figure 2a). Next, each rectangle was assigned suitable material properties, such as air, skin, fat, and bone. In this work, the bulk of the body was modeled as a conductor. For the implementation, the edges of the rectangles corresponding to the bulk region were configured as floating conductors. In addition, the edges corresponding to the electrodes: Tx-Signal, Tx-GND, Rx-Signal, and Rx-GND were also configured as conductor. A voltage of 5 V was applied to the Tx-Signal electrode and 0 V to the Tx-GND electrode. The Rx-Signal and Rx-GND electrodes were left floating. A boundary region was defined by constructing a rectangular region approximately 10 times larger than the model dimensions. The edges of this rectangle were set to a potential of 0 V to ensure realistic decay of the electric field with increasing distance from the electrodes.

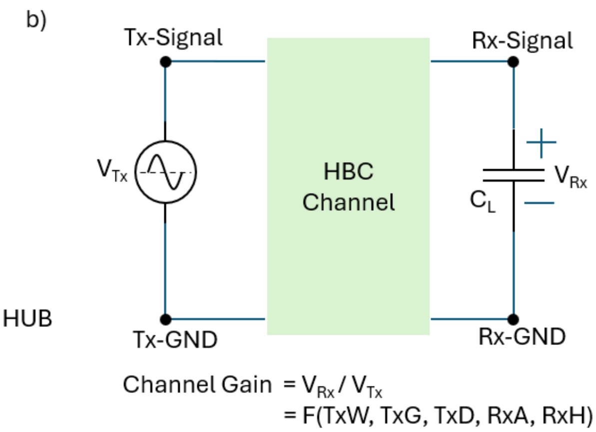
Simulations were setup to generate the voltage distribution over the modeled region. Additionally, the simulations were set up to measure the voltage difference between the Rx-Signal and Rx-GND electrodes, representing the receiver-side signal strength. The model parameters, TxD, TxW, TxG, RxH, and RxA, were swept in the script. For each parameter setting, the model was reconstructed, and the simulation was executed to compute the corresponding receiver-side signal strength and thus the channel gain [9], [15].

B. Demonstration of FEMM Simulation

A representative simulation was executed using a fixed set of parameters to evaluate the modeling methodology. In this simulation, the transmitting electrode was positioned 20 cm from the body edge (TxD), with a width of 3 cm (TxW) and a separation of 3 cm (TxG) between the signal and ground electrodes. The receiving electrode, Rx-GND with a vertical height of 30 cm (RxH) was placed. The electrode was seperated from Rx Signal electrode by 10 cm(RxA). The thicknesses of both the Tx and Rx electrodes were 0.2 cm.

Figure 3 presents the FEMM output, showing the voltage density plots across the body and the surrounding medium. The plot shows varying potential gradients throughout the HBC channel. Under these conditions, the receiver-side signal strength measured between the Rx-Signal and Rx-Ground electrodes was approximately 0.1366 V, resulting in a channel gain of approximately –31.26 dB. The observed voltage across





c)

Material	Thickness	Relative Permittivity
Skin	0.063 cm	2e4
Fat	0.457 cm	1e2
Muscle	1.255 cm	8e3
Bone Marrow	0.373 cm	2.5e2
Bone	0.352 cm	2.5e2
Wood	-	2

Fig. 2. (a) Simulation model developed for the galvanic HBC configuration shown in Figure 1. The model is constructed using geometries and materials representing the anatomy of the human body, along with the transmitting and receiving electrodes and the table. (b) Simplified circuit schematic of the HBC channel. (c) Material properties used for modelling the arm and the wooden table.

the receiving electrodes confirms the signal transmission from the transmitter to the receiver via the body.

C. Channel Gain vs. Model Parameters

In the previous experiment, model parameters were fixed. In this experiment, we varied the values of the model parameters within a defined range to study their impact on the receiver-side signal strength and, consequently, the channel gain.

The simulation was set up as follows. The parameters TxD, TxW, TxG, RxH, and RxA were varied. TxD was varied from 10 cm to 40 cm in steps of 5 cm. TxW was varied from 1 cm to 10 cm in steps of 1 cm. TxG was varied from 1 cm to 10 cm in steps of 1 cm. RxH was varied from 10 cm to 50 cm in steps of 10 cm, and RxA from 5 cm to 40 cm in steps of 5 cm. In each simulation set, only one parameter was varied while the others were held constant at their nominal values. The nominal values used were: TxD = 20 cm, TxW = 3 cm, TxG = 3 cm, RxH = 30 cm, and RxA = 10 cm. The results of these simulations are presented in Figure 4, which plots the received signal strength, V_{Rx}, as a function of the model parameters. We justify these findings in the following paragraphs with the help of the simplified circuit model and channel gain equation in Figure 5.

Figure 4a) shows that increasing TxD leads to an increase in Rx voltage for the following reason. At higher TxD values, more electric field lines couple to the receiver through the body, thereby increasing the coupling capacitance between Tx-GND and Rx-GND electrode, C_R, and thus the channel gain.

In Figure 4b), the Rx voltage initially decreases with increasing TxW, followed by an increase at greater widths for the following reason. Initially, increasing TxW causes higher field concentration between the transmitting electrode reducing field couplings to receiving electrodes, resulting in lower C_F and C_R. However, beyond a certain value of TxW, the increased surface area of the transmitting electrodes increases the coupling of fields with the receiver via the body and air, increasing C_F and C_R and thus the channel gain.

Figure 4c) shows that an increase in TxG results in a decline in Rx voltage. At lower values of TxG, field lines are concentrated between the Tx electrodes, resulting in lower coupling to receiving electrodes and thus lower values of C_F and C_R. Lower values of these capacitors result in lower gain. But as the TxG increases, a wider separation is created between the Tx-Signal and Tx-GND electrodes, which reduces the strength of the electric field between them. Field lines are more dispersed now, resulting in more coupling to the receiving electrodes, resulting in higher C_F and C_R and thus the channel gain.

Figure 4d) shows that increasing RxH results in higher Rx voltage due to the following reason. Larger Rx electrodes intercept more electric field lines from the trasnmitting electrodes, increasing C_R and, consequently, the channel gain.

Figure 4e) shows that increasing RxA results in higher Rx voltage due to the following reason. As the RxA increases, separation between receiving electrodes also increases resulting in lower C_L, increasing the channel gain.

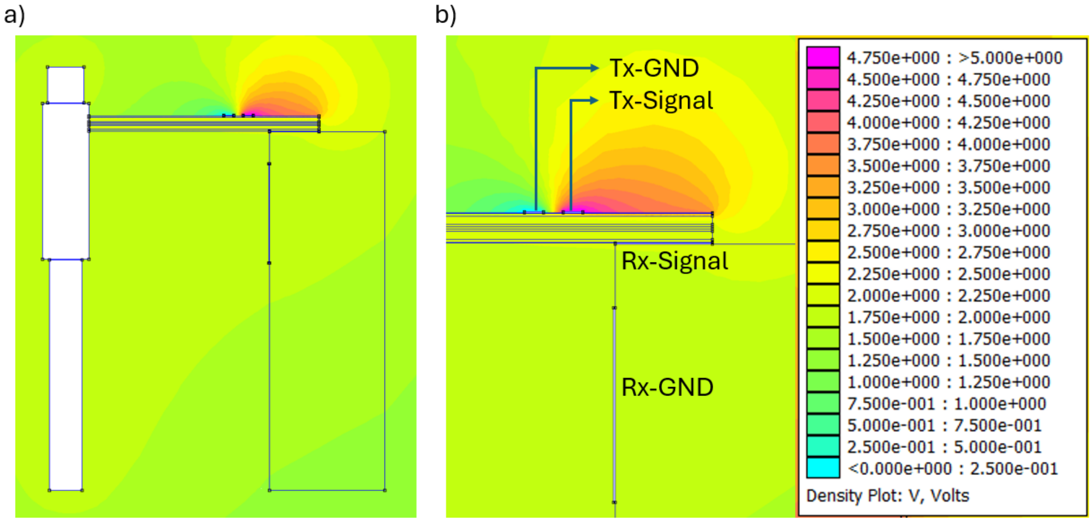


Fig. 3. Simulation results: a) Electric potential distribution for the galvanic HBC configuration. b) Zoomed-in view of (a) highlighting the Tx-Signal, Tx-GND, Rx-Signal, and Rx-GND electrodes.

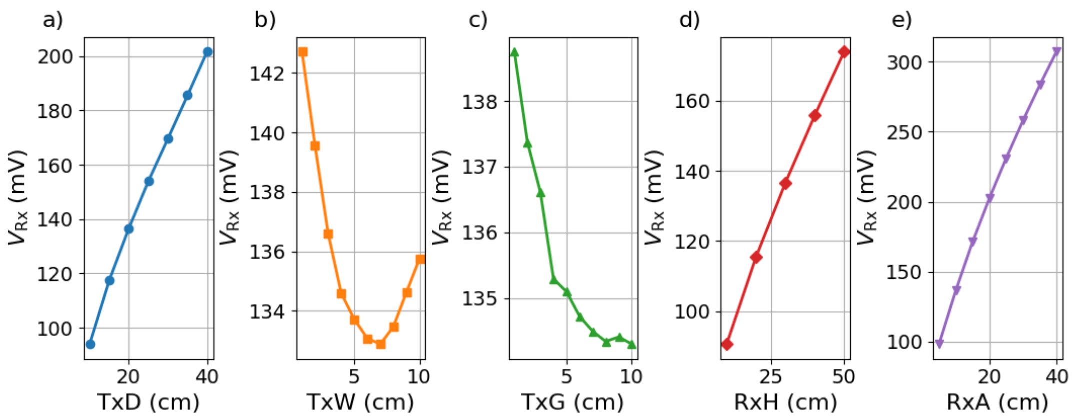


Fig. 4. Simulation results showing the effect of model parameters TxG, TxW, TxD, RxH, and RxA on the received voltage, V_{Rx}.

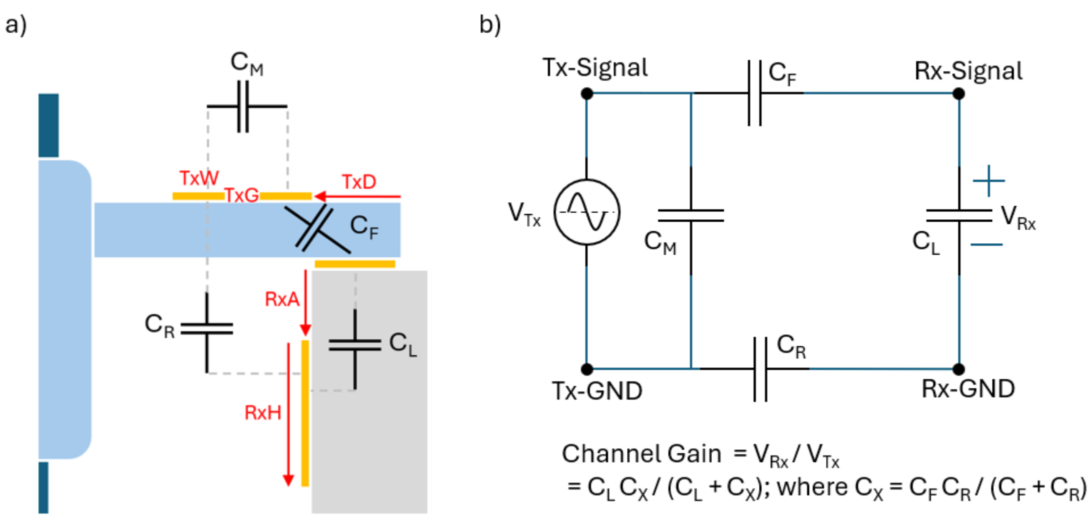


Fig. 5. a) Highlighting coupling capacitances of the HBC configuration shown in Figure 2. b) Corresponding equivalent low-frequency circuit model and channel gain expression.

V. Conclusion

Several studies have focused on simulating HBC for wearable-to-wearable communication. However, to the best of our knowledge, no simulation studies have investigated an HBC configuration consisting of a wearable transmitter and a grounded receiver. Here, grounded refers to a receiver that is fixed to a grounded object, such as a table. To address this gap, this work presents a simulation-driven approach to investigate a galvanic Human Body Communication (HBC) channel with a wrist-worn transmitter and a grounded receiver.

A simplified two-dimensional human body model was constructed for simulation purposes. The body comprised key components including the head, torso, and legs. The arm region, which is in the critical path of signal transmission, was modeled in detail using layers of skin, fat, muscle, bone, and bone marrow. Electrodes for the transmitter were mounted on the wrist, while receiving electrodes were placed on a table. The Finite Element Method (FEM) simulator, FEMM, was used to run the simulations. These simulations were carried out to study the effect of electrode size and position on the HBC channel. The results indicate that electrode geometry and positioning are critical factors in determining HBC performance. Channel gain was found to depend on model parameters including TxW (width of the Tx electrode), TxG (gap between Tx electrodes), TxD (distance from the Tx electrodes to the arm end), RxA (alignment offset of the Rx GND electrode), and RxH (vertical height of the Rx GND electrode). Increasing TxD, RxH, and RxA resulted in stronger signal coupling, leading to higher Rx voltage and, therefore, higher channel gain. In contrast, a larger TxG reduced coupling efficiency due to weakened field concentration. Increasing TxW initially reduced coupling, but beyond a certain point, the coupling magnitude increased.

Our findings can help in the design of compact wearables and receivers for more reliable body-coupled communication. As part of future work, we intend to extend the simulation framework to support additional variants of the HBC configuration with grounded receivers discussed in this work.

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